Enea Manzi

Curriculum Vitae

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Education

09/2024 - 07/2026 **Master's Degree in Cyber Security**, *University of Milan*, (Milan, IT)
Thesis: "Design and Implementation of an API-Agnostic Tool for REST API Security Assurance"
Supervisors: Marco Anisetti, Claudio Agostino Ardagna
Expected Graduation: 07/2026

09/2021 - 10/2024 **Bachelor's Degree in System and Network Security**, *University of Milan*, (Milan, IT)
Thesis: "Design and Implementation of a Framework for Evaluating Distributed Systems"
Supervisors: Marco Anisetti, Filippo Berto
Final Grade: 110/110 cum laude

09/2016 - 06/2021 **High School Diploma in Computer Science**, *IIS Badoni*, (Lecco, IT)
Final Grade: 93/100

Work Experience

12/2025 - 07/2026 **Curricular Internship for Master's Thesis**, *SESAR Lab, University of Milan*, (Milan, IT)
Secure Service-oriented Architectures Research Lab (SESAR Lab)
Designed and implemented **APIGuard Assurance**, an **API-agnostic**, **contract-driven** tool for automated REST API **security assurance** based on a taxonomy of eight domains and *specified oracles*, validated against a real-world target.

03/2024 - 09/2024 **Curricular Internship for Bachelor's Thesis**, *SESAR Lab, University of Milan*, (Milan, IT)
Secure Service-oriented Architectures Research Lab (SESAR Lab)
Contributed to the design and implementation of an **Assurance Engine** for the verification of **non-functional properties (NFP)** in distributed systems, built in **Rust** and deployed on **Kubernetes**, developing a new connector to integrate **Elasticsearch** as a monitoring backend.

01/2020 - 02/2020 **WBL: web developer**, *DaTrik Solutions*, (Casargo, IT)
High school work-based-learning
Developed a web application using HTML, JavaScript, and jQuery as part of a high school work-based learning program (PCTO).

Research Activities

12/2025 - 07/2026 **SESAR Lab, University of Milan**, (Milan, IT)

Secure Service-oriented Architectures Research Lab (SESAR Lab)

- Analyzed existing approaches to REST API testing, observing that most focus on **functional** and **syntactic** conformance, while **security violations** remain comparatively unaddressed.
- Defined a taxonomy of **eight security domains** to structure systematic assessment coverage.
- Proposed an **API-agnostic**, **contract-driven** methodology, based on *specified oracles*, that derives target knowledge exclusively from the **OpenAPI Specification (OAS)** and a deployment configuration file.
- Addressed the *oracle problem* in automated security testing by deriving, for each security control, an evaluation criterion from the specific knowledge and expectations of the test being performed, traceable to **OWASP** categories and **CWE** codes.
- Achieved **deterministic**, **reproducible** results, a prerequisite for assurance integrated into continuous development, through a **DAG**-based scheduling approach exposing **semantic exit codes** for **CI/CD** integration.
- Validated the methodology against a real-world target, **Forgejo** exposed through the **Kong** API gateway, confirming correct detection of security violations and deterministic, reproducible results.

03/2024 - 09/2024 **SESAR Lab, University of Milan**, (Milan, IT)

Secure Service-oriented Architectures Research Lab (SESAR Lab)

- Investigated existing approaches to verifying and monitoring **non-functional properties (NFP)** in distributed systems.
- Contributed to the development of an **Assurance Engine** for distributed systems, building a new connector to integrate **Elasticsearch** as a monitoring backend.
- Formalized contracts to jointly evaluate multiple non-functional properties (**CIA triad**, reliability, scalability, performance), validated experimentally on a **Kubernetes** testbed with minimal computational overhead.

Projects

APIGuard Assurance *Master's Thesis Project (A.Y. 2025/2026)*

Development of an API-agnostic, contract-driven tool for REST API security assurance, with a particular focus on semantic security violations that traditional functional and syntactic testing does not capture.

- Built in **Python 3.12**, with a **dynamic test-discovery mechanism** allowing new security checks to be added without modifying the core engine.
- Implemented *specified oracles* directly in each test's code: every check encodes its own evaluation criterion, derived from the domain knowledge of the control under test.
- Enforced **strict static typing** (mypy) and **runtime validation** (Pydantic), with full documentation coverage across the public API.
- Built **reproducible packages** with **Hatch**, verifying identical SHA-256 hashes across repeated builds.
- Designed an **extensible** connector architecture to integrate external security scanners, decoupling test logic from tool-specific output parsing.

Event-Driven IoT Architecture (Kafka + Kong) *Cloud Computing Technologies Course Project (A.Y. 2025/2026)*
Development of a cloud-native microservices infrastructure on Kubernetes for IoT ingestion and monitoring, with a particular focus on **non-functional properties** such as scalability, fault tolerance, and edge-to-core security.
- Microservices architecture based on **Apache Kafka** in KRaft mode (ZooKeeper-less) for asynchronous decoupling and reliable message handling (*Zero Data Loss*).
- Integration of **Kong API Gateway** to implement the *Gateway Offloading* pattern, centralizing Authentication (API Key), Rate Limiting, and Load Balancing at the edge.
- Storage layer optimized on **MongoDB** using **Time Series Collections** and hybrid compression (LZ4/Zstd) to maximize storage efficiency and analytical performance.
- Full **Kubernetes** orchestration managed via **Helm**, featuring automatic scalability (HPA) and advanced security (Secrets, TLS/SASL) to meet non-functional requirements.

FTP Client and Server *Computer Networks Course Project (A.Y. 2022-2023)*
Developed a multi-threaded **FTP-compliant** client-server application in **C** for file transfer, fully interoperable with **FileZilla**.
- Architecture: Multi-threaded server based on **Socket Programming** for concurrent session management.
- Data Modes: Native support for both Active (PORT) and Passive (PASV) data transfer modes.
- Protocol Management: Implemented essential *FTP commands* (e.g., USER, PASS, LIST, RETR, STOR, CWD...)
- **Key skills: C**, **socket programming**, **multithreading**, **network protocols**, **FTP**, **Git**.

Exhibition Web App *Web and Mobile Programming Course Project (A.Y. 2021-2022)*
Built a dynamic web application serving as a digital portfolio for displaying multimedia content.
- Developed the backend with *Node.js/Express* and the frontend with *HTML5*, *CSS3*, *JavaScript/jQuery*.
- Integrated *MongoDB* for document-oriented storage and *AJAX* for asynchronous client-server communication (GET, POST, DELETE).
- Managed source code and collaboration using **Git** and **GitHub**.
- **Key skills:** *Node.js*, *Express*, *MongoDB*, *AJAX*, *REST APIs*, *jQuery*, **Git**.

Publications

Jun 2026 **An API-Agnostic, Contract-Driven Approach for REST API Security Assurance**
Enea Manzi, Marco Anisetti, Claudio A. Ardagna
[Submitted to ITADATA 2026]
Automated REST API testing has traditionally focused on functional compliance, leaving security assurance comparatively underexplored; existing tools operate at the syntactic level and cannot reliably detect the semantic violations behind most OWASP API Security Top 10 categories. This work proposes a contract-driven approach that derives all target knowledge from the OpenAPI specification and a configuration file, embedding evaluation criteria directly into test logic and organizing checks into an eight-domain taxonomy, executed through a deterministic DAG-based engine. Validated on Forgejo behind Kong, the approach produced reproducible results and detected a real discrepancy between the declared specification and actual system behavior, demonstrating an assessment process directly integrable into continuous development pipelines.

Languages

Italian Mother tongue

English Intermediate *Listening B2 - Reading B2 - Writing B2 - Spoken Production B2 - Spoken Interaction B2*

Skills

Programming Languages	*C*, *Java*, *Python*, *SQL*, Web (*HTML*, *JavaScript/Node.js*), OOP Concepts, low-level programming
Database	*MySQL*, Database Design, Implementation and Adminsitration
Architectures & Software	*Docker*, *Kubernetes*, microservices architectures, REST API, Git, Kafka, Kong API Gateway, CI/CD
Operating Systems	Fresh OS installation, Linux Administration and Shell, Windows
Networking & Security	Network Protocols, ICT Network Security Risk, Secure Coding Practices, API Security, Continuous Security Assurance
Tools	Visual Studio Code, GitHub

Soft Skills

Thinking	analytically, critically, proactively
Problems	address problems critically, solve problems/create solutions
Communication	cooperate with colleagues, work in teams, manage teamwork
Others	demonstrate curiosity, cope with stress, prioritise tasks

Bellano, June 26, 2026

Enea Manzi